BUSINESS BLUEPRINT

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OVERVIEW

Glossary

Term	Definition	Note
ABAP	Advanced Business Application Programming — SAP's proprietary language	
ALV	ABAP List Viewer — SAP's standard grid/list display component	
BCS	Business Communication Services — SAP API for sending emails	
Bug	A defect or issue found during software testing	
BUG_ID	Unique identifier for a bug record, format: BUG0000001 (auto-generated)	
Custom Control	A region on a Dynpro screen reserved for GUI controls (ALV, TextEdit)	
DEV	Developer — role responsible for fixing bugs	
Dynpro	Dynamic Program — SAP screen technology (also called Dynscreen)	
Evidence	Proof file (Excel .xlsx) uploaded to confirm bug report, fix, or verify	
F4 Help	SAP search help — dropdown/popup for field value selection	
Final Testing	Status 6 — bug has been fixed, Final Tester is verifying	
GOS	Generic Object Services — SAP framework for attaching files to records	
GUI Status	SAP toolbar/menu definition for a screen (created in SE41)	
In Progress	Status 3 — Developer is actively working on the bug	
MGR	Manager — role with full system access	
Module Pool	Type M ABAP program — the basis of complex SAP screen applications	
New	Status 1 — bug just created, not yet assigned	

Package	`ZBUGTRACK` — SAP development package grouping all objects	
PAI	Process After Input — screen event triggered after user action	
PBO	Process Before Output — screen event triggered before display	
Pending	Status `4` — temporarily blocked, awaiting information	
Project	A development project that groups related bugs (table: ZBUG_PROJECT)	
Rejected	Status `R` — Developer rejected the bug assignment	
Resolved	Status `V` — Final Tester confirmed fix; terminal state	
SE11	SAP Data Dictionary — tool for creating tables, domains, data elements	
SE38/SE80	SAP ABAP Workbench — code editor	
SE41	SAP Menu Painter — tool for creating GUI Statuses and Title Bars	
SE51	SAP Screen Painter — tool for creating Dynpro screens	
SE93	SAP Transaction Maintenance — for creating/editing T-codes	
SmartForm	SAP form printing tool — used for PDF reports and email body	
T-Code	Transaction Code — SAP shortcut to launch a program (e.g., ZBUG_WS)	
Tester	Role responsible for reporting bugs and verifying fixes	
Waiting	Status `W` — no suitable Dev/Tester found; awaiting Manager assignment	
Workload	Number of active bugs (status 2/3/4/6) assigned to a user	
ZBUG_WS	The T-Code entry point for the Bug Tracking system	

Flow Chart Shapes Usage

Shape	Usage / Meaning

Rectangle (solid border)	Process step / System action (e.g., Auto-assign, Save record)
Diamond	Decision / Condition branch (e.g., Bug Type = Code or Config?)
Rounded Rectangle (oval)	Start / End terminal state
Parallelogram	Input from user / Output to user
Dashed arrow	Conditional flow (only under certain conditions)
Solid arrow	Normal sequential flow
Bold border rectangle	System-automated step (no user interaction required)

ORGANIZATIONAL STRUCTURE

The system operates within **SAP System S40, Client 324** and is accessed through T-Code **ZBUG_WS**. Three distinct roles exist within the system, each mapped to a record in table **ZBUG_USERS** (field `ROLE`, CHAR 1).

System Roles

Code	Role	Description	SAP Account (Demo)
`M`	**Manager**	Full system access. Manages projects, assigns bugs, closes bugs, views dashboard statistics, manages user accounts.	`DEV-089`
`D`	**Developer**	Receives assigned bugs. Updates bug status (InProgress, Fixed, Pending, Rejected). Uploads fix evidence. Can only act on bugs where `DEV_ID = SY-UNAME`.	`DEV-061`
`T`	**Tester**	Creates bugs, uploads bug report evidence. Verifies fixed bugs (Final Testing phase). Can also self-fix Config-type bugs as both Developer and Tester.	`DEV-118`

Role Determination at Runtime

Role is resolved from table `ZBUG_USERS` on login:

```
SELECT SINGLE role FROM zbug_users INTO @gv_role
  WHERE user_id = @sy-uname
    AND is_del  <> 'X'
    AND is_active = 'X'.
```

Project-Level Role Assignment

Each user can hold a different role per project, stored in `ZBUG_USER_PROJEC` (Key: MANDT + USER_ID + PROJECT_ID, field `ROLE`). This allows cross-project flexibility (e.g., a user may be Manager in Project A and Developer in Project B).

Permission Matrix Summary

Action	M	D	T
Create Bug	✓	—	✓
Delete Bug	✓	—	—
Change Status (via Popup 0370)	✓	✓	✓
Assign / Reassign Bug	✓	—	—
Upload Fix Evidence	✓	✓	—
Upload Report Evidence	✓	—	✓
Upload Verify Evidence	✓	—	✓
View All Bugs	✓	—	—
Create / Edit Project	✓	—	—
Add / Remove Project Users	✓	—	—
View Dashboard Statistics	✓	—	—
Download Templates	✓	—	✓
Print SmartForm	✓	✓	✓
Send Email	✓	✓	✓

BUSINESS PROCESS

BP-BUG-01: Bug Lifecycle Management

Process Flow

```
┌──────────────────────────────────────────────────────────────────────────┐
│  Tester creates Bug on Screen 0300 (from Project context)                 │
│  STATUS = 1 (New)  │  BUG_TYPE = Code (C) or Config (F)                  │
└──────────────────────────────────────────────────────────────────────────┘
                                                          │

┌──────────────────────────────▼──────────────────────────────┐
                            │                  BUG_TYPE = Code (C)?
│
└──────────────────────────┬──────────────────────────┬──────────────────┘
                                                     │ Yes                 │ No
(Config F)
                                                     │                     │
                      ┌──────────────────────────────▼──┐
┌──────────────▼──────────────────┐
                      │  Auto-Assign (Phase A)        │    │  Tester self-fixes
Config bug │
                      │  Find Dev: same module,       │    │  DEV_ID = SY-UNAME
│
                      │  workload < 5, Available      │    │  STATUS:
1→2→3→5→6→V           │
                      └──────────────────────────────┘
└──────────────────────────────┘
                         │ Dev found      │ No Dev
                         │               │
                  ┌──────▼──────┐   ┌──────▼──────┐
                  │ STATUS = 2  │   │ STATUS = W  │
                  │ (Assigned)  │   │ (Waiting)   │◄── Manager assigns manually
                  └─────────────┘   └─────────────┘
                         │
                         │
                  Developer starts: STATUS = 3 (In Progress)
                         │
               ┌─────────┴──────────────────────┐
               │ Dev may Pending (4) or         │
```

```
              │ Reject (R) with note          │
              └──────────────┬────────────────┘
                             │
                             │ Dev uploads fix evidence
                             │ STATUS = 5 (Fixed)
                             │
                  ┌──────────▼──────────────────────────────────────┐
                  │  Auto-Assign (Phase B)                          │
                  │  Find Tester: same module, workload < 5         │
                  └──────────┬──────────────────────────┬──────────┘
                             │ Tester found             │ No Tester
                             │                          │
                  ┌──────────▼──────────┐     ┌──────────▼──────────┐
                  │ STATUS = 6          │     │ STATUS = W          │◄── Manager assigns
Final Tester
                  │(Final Testing)      │     │  (Waiting)          │
                  └──────────┬──────────┘     └─────────────────────┘
                             │
                             │
              ┌──────────────┴──────────────┐
              │ Test PASS                   │ Test FAIL
              │                             │
      ┌───────▼──────────┐          STATUS = 3 (back to Dev)
      │ STATUS = V       │          TRANS_NOTE mandatory
      │  (Resolved)      │
      │  TERMINAL        │
      └──────────────────┘
```

Process Description

Step #	Step Name	Detailed Description	Role
1	Create Bug	Tester navigates Screen 0300 (from a project context). Fills Title, SAP Module, Priority, Severity, Bug Type. BUG_ID is auto-generated. STATUS auto-set to 1 (New). TESTER_ID auto-set to SY-UNAME.	Tester / Manager
2	Auto-Assign Developer	System queries ZBUG_USER_PROJEC for Developers in the same project + module. Selects the one with workload < 5 and lowest active bugs. Sets STATUS → 2 (Assigned), DEV_ID	System (auto)

		= selected user. If none found → STATUS = `W` (Waiting), Manager notified by email.	
3	Manual Assignment	If STATUS = `W`, Manager opens Status Transition Popup (Screen 0370), selects Developer manually, sets STATUS → `2`. Manager can also directly assign from any state.	Manager
4	Start Working	Developer opens popup 0370, transitions STATUS → `3` (In Progress). Developer may set `4` (Pending) if blocked.	Developer
5	Fix Bug	Developer uploads fix evidence (fix_report.xlsx via ZBUG_EVIDENCE). Status must have at least 1 evidence file. Transitions STATUS → `5` (Fixed) via popup 0370.	Developer
6	Auto-Assign Final Tester	System queries ZBUG_USER_PROJEC for Testers in same project + module. Workload = COUNT bugs WHERE status = `6`. Selects lowest-workload Tester < 5. Sets STATUS → `6`, VERIFY_TESTER_ID = selected tester. If none → STATUS = `W`.	System (auto)
7	Final Testing	Assigned Tester verifies the fix. Opens popup 0370. **Pass:** enters TRANS_NOTE → STATUS = `V` (Resolved). **Fail:** enters TRANS_NOTE → STATUS = `3` (back to InProgress).	Tester (Final)
8	Resolved	STATUS = `V` is the terminal state. No further transitions allowed. Bug is considered complete.	—
9	Rejection Flow	Developer may reject (`R`) with mandatory TRANS_NOTE. Manager reassigns to another Developer → STATUS = `2`. This is logged in ZBUG_HISTORY with Action = `RS`.	Developer / Manager

BP-BUG-02: Status Transition Popup (Screen 0370)

Process Flow

```
User clicks "Change Status" button on Screen 0300 (Bug Detail)
  |
  ▼
```

```
Screen 0370 opens as Modal Dialog
  │
  ├── Read-only fields shown:
  │     BUG_ID, TITLE, REPORTER (TESTER_ID), CURRENT_STATUS
  │
  ├── Input fields enabled based on CURRENT_STATUS:
  │
  │   STATUS = 1 (New):
  │     NEW_STATUS dropdown: [2-Assigned, W-Waiting]
  │     DEVELOPER_ID: OPEN (mandatory if → 2)
  │
  │   STATUS = 2 (Assigned):
  │     NEW_STATUS dropdown: [3-InProgress, R-Rejected]
  │     TRANS_NOTE: mandatory if → R
  │
  │   STATUS = 3 (In Progress):
  │     NEW_STATUS dropdown: [5-Fixed, 4-Pending, R-Rejected]
  │     TRANS_NOTE: open (mandatory if → R)
  │     BTN_UPLOAD: open (mandatory if → 5, evidence required)
  │
  │   STATUS = 4 (Pending):
  │     NEW_STATUS dropdown: [2-Assigned]
  │     DEVELOPER_ID: open (can change Dev)
  │
  │   STATUS = 6 (Final Testing):
  │     NEW_STATUS dropdown: [V-Resolved, 3-InProgress]
  │     TRANS_NOTE: mandatory (test result note)
  │
  ▼
User clicks CONFIRM → System validates → Updates ZBUG_TRACKER
  → Logs ZBUG_HISTORY (Action: ST)
  → Auto-Assign triggered if applicable
  → Email notification sent
```

Process Description

Step #	Step Name	Detailed Description	Role
1	Open Popup	User clicks STATUS_CHG button on Screen 0300. CALL SCREEN 0370 STARTING AT	Developer / Tester / Manager

		col row. Screen fields populated from current ZBUG_TRACKER record.	
2	Select Transition	User selects target status from dropdown (NEW_STATUS). Available options depend on current status and user role. System enforces transition matrix.	Developer / Tester / Manager
3	Fill Required Fields	Depending on transition: fill DEVELOPER_ID, FINAL_TESTER_ID (via F4), or TRANS_NOTE (free text). Upload button available for evidence when transitioning to Fixed (5).	Developer / Tester / Manager
4	Confirm	User clicks CONFIRM. System validates: permission check, mandatory fields, evidence check (for → 5), TRANS_NOTE check (for → R, → V, → 3-from-6). If valid → update DB + log + email.	System
5	Auto-Assign Trigger	If transitioning to Fixed (5): system auto-assigns Final Tester. If → Assigned (2) from status 1/W/4: system may auto-assign or use DEVELOPER_ID from popup.	System (auto)

BP-PRJ-01: Project Management

Process Flow

```
Manager creates Project on Screen 0500
  │  Fills: PROJECT_ID, PROJECT_NAME, DESCRIPTION, START_DATE, END_DATE
  │  STATUS = 1 (Opening)
  │
  ▼
Manager adds Users to Project
  │  Screen 0500, Table Control TC_USERS
  │  Each user assigned Role: M/D/T in ZBUG_USER_PROJEC
  │
  ▼
Project transitions to STATUS = 2 (In Process) — Manager
  │
  │  [Bugs are created, tracked, fixed, resolved within the project]
```

```
  |
  ▼
Project transitions to STATUS = 3 (Done) — Manager
  |  CONDITION: All bugs in project must be Resolved (V) or Closed (7)
  |  System validates: COUNT open bugs = 0
  |
  OR
  ▼
Project transitions to STATUS = 4 (Cancelled) — Manager
```

Process Description

Step #	Step Name	Detailed Description	Role
1	Create Project	Manager opens Screen 0500 (from Project List 0400, action: CREA_PRJ). Fills PROJECT_ID (CHAR 20), PROJECT_NAME, DESCRIPTION, START_DATE, END_DATE, NOTE. STATUS auto-set to 1 (Opening).	Manager
2	Assign Team	On Screen 0500, Table Control TC_USERS shows project members. Manager uses ADD_USER button to add SAP users with role M/D/T. Uses REMO_USR to remove.	Manager
3	Activate Project	Manager changes status → 2 (In Process). Bugs can now be created under this project.	Manager
4	Search Projects	Screen 0410 (Project Search) is the initial screen. User fills filters (Project ID, Name, Status) and clicks Execute → Screen 0400 (Project List ALV).	All Roles
5	Upload Projects	Manager uploads Excel file (template: ZTEMPLATE_PROJECT via SMW0). System parses via TEXT_CONVERT_XLS_TO_SAP, validates, and batch-inserts into ZBUG_PROJECT.	Manager
6	Close Project	Manager sets STATUS → 3 (Done). System checks: all bugs must be Resolved (V). If any	Manager

		open bug exists → error message, transition blocked.	

REPORTS

Based on system requirements, the following reports and outputs are available within the SAP Bug Tracking Management System.

No.	Description	Detail	T-Code / Access
1	Bug List ALV Report	Full list of bugs per project with color-coded status, priority, module, tester, developer. Supports filter, sort, export to Excel. Role-based: Tester sees own bugs, Developer sees assigned bugs, Manager sees all.	ZBUG_WS → Screen 0200
2	Bug Detail SmartForm (PDF)	Printable PDF report for a selected bug. Includes: Bug ID, Title, Module, Priority, Status, Severity, Type, Tester, Developer, Created/Closed dates, history log, notes. Triggered by PRINT button on Screen 0200.	ZBUG_WS → Screen 0200 → PRINT
3	Project List ALV Report	List of all projects accessible to the current user. Shows Project ID, Name, Status, Manager, Start/End date, user count. Filtered from Screen 0410 (Project Search).	ZBUG_WS → Screen 0400
4	Bug History Log	Complete audit trail for a single bug. Shows all status changes, assignments, uploads, field edits with timestamp, user, old value, new value, and reason. Readonly ALV on Tab 0360 of Bug Detail.	ZBUG_WS → Screen 0300 → Tab History
5	Bug Search Results	Ad-hoc search report across all bugs in a project. Supports multi-field filter (Bug ID, Title, Status, Priority, Module, Tester, Developer). Output on Screen 0220.	ZBUG_WS → Screen 0200 → SEARCH
6	Dashboard Statistics (Manager only)	Real-time summary header on Screen 0200 showing: Total bugs, by Status (New/Assigned/InProgress/Fixed/FinalTesting/Wait-	ZBUG_WS → Screen 0200 (Manager)

		ing/Resolved/Rejected), by Priority (H/M/L), by Module.	
7	Email Notification Log	Automatic email sent on key events: Bug Created, Assigned, Status Changed, Rejected, Resolved. Sent via SAP BCS API (CL_BCS). Viewable in SOST.	SOST (SAP standard)
8	Evidence Files (ZBUG_EVIDENCE)	Binary evidence files stored in custom table ZBUG_EVIDENCE. Three types per bug: Bug_report.xlsx (Tester), fix_report.xlsx (Developer), confirm_report.xlsx (Final Tester). Downloadable from Screen 0300 Evidence tab.	ZBUG_WS → Screen 0300 → Tab Evidence